

Project Documentation: Titanic EDA & Feature Engineering

1. Overview

This document outlines the exploratory data analysis (EDA), data cleaning, and feature engineering steps performed on the Titanic dataset to prepare it for machine learning classification models.

2. Exploratory Data Analysis (EDA) Insights

Based on statistical evaluations and visualizations, the following key patterns were uncovered:

- **Uniform Identifier:** The PassengerId distribution plot demonstrates a completely flat, uniform distribution, confirming it is merely a row index with zero correlation (-0.01) to survival.
- **Class Impact:** Passenger class (Pclass) shows the strongest negative correlation (-0.34) with survival, indicating that higher-tier classes (1st class) experienced significantly better survival rates.
- **Fare Influence:** Ticket Fare exhibits a positive correlation (0.26) with survival, showing that higher-paying passengers had better survival outcomes.
- **Economic Link:** A strong negative correlation (-0.55) exists between Pclass and Fare, reflecting that premium classes naturally commanded higher ticket prices.
- **Family Dynamics:** SibSp (siblings/spouses) and Parch (parents/children) share a positive correlation (0.41), showing that families often traveled together.

3. Feature Engineering & Transformations

- **Feature Dropping:** PassengerId was removed from the feature set because its uniform distribution and lack of correlation offer no predictive power.
- **Categorical Encoding:** Text-based categorical columns (such as Sex and Embarked) were converted into numerical formats using One-Hot Encoding (`pd.get_dummies`) to ensure compatibility with machine learning algorithms.
- **Feature Creation:** Combined SibSp and Parch into a single consolidated feature called FamilySize to better capture total family presence onboard.
- **Feature Scaling:** Continuous numerical features (like Age and Fare) were standardized using Scikit-learn's StandardScaler to bring them onto a uniform scale.

4. Final Selected Features for Modeling

- Pclass (Passenger Class)
- Sex (Encoded Gender)
- Age (Scaled)

- Fare (Scaled)
- FamilySize (Engineered Feature)
- Embarked (Encoded Port of Embarkation)